Indirect reciprocity with inertial reputation updating

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Abstract

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Introduction

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Results

A framework for inertial reputation updating

We consider indirect reciprocity in a well-mixed population playing donation games, in which a donor can pay a cost c to give a recipient a benefit $b > c$. Unconditional cooperators (ALLC) intend to help every recipient, unconditional defectors (ALLD) never intend to help, and discriminators (DISC) intend to help recipients they regard as good. Reputations are private: every individual keeps its own record of every other individual, and an observer judges a donor's realized action by applying a shared social norm to the observer's own view of the recipient. We study the four second-order norms Scoring (SC), Shunning (SH), Stern Judging (SJ) and Simple Standing (SS). They agree on actions towards recipients regarded as good and differ in how they judge cooperation and defection towards recipients regarded as bad. Intended cooperation fails with probability e_1, and an observer records the opposite of the prescribed assessment with probability e_2 (Methods).

In the standard model, every new assessment replaces the observer's previous binary image of the donor: one negative assessment turns a good image bad, and one positive assessment turns a bad image good (Fig. 1a). We instead let observers accumulate evidence before they reverse an established judgement (Fig. 1b). Negative assessments move a record in good standing through k_1 warning states before it becomes bad, and positive assessments move a record in bad standing through k_2 recovery states before it becomes good. Evidence consistent with the established judgement resets the opposing path. Losing good standing therefore requires k_1+1 consecutive negative assessments of the same individual by the same observer, and regaining it requires k_2+1 consecutive positive ones. Deterioration and recovery are thus two separately tunable evidence-accumulation processes. We refer to k_1 as tolerance depth and to k_2 as strictness depth.

Intermediate states must still be translated into treatment. When a record is consulted, for choosing an action or for assessing someone else's action, it is read out as a binary image: an individual in warning state W_m is still treated as good with probability α^m, and an individual in recovery state S_n is still treated as bad with probability β^n. The depths k_1 and k_2 thus determine how much consistent evidence completes a reversal, whereas the readout parameters α and β determine how strongly incomplete evidence already affects social treatment. With full readout, $\alpha=\beta=1$, the established image persists until a path is completed; with $\alpha=\beta=0$, or with $k_1=k_2=0$, updating is instantaneous. Throughout, we compare three settings: no inertia; full one-step inertia, $(k_1,k_2)=(1,1)$ with $\alpha=\beta=1$; and norm-specific multi-step inertia with full readout, introduced below. We first ask how this temporal integration of behavioural evidence changes reputation formation itself.

Directional inertia reshapes private reputation dynamics

Private assessment is a central obstacle to indirect reciprocity, because observers who hold different views of the same recipient reach different verdicts about the same action, and norms that sustain cooperation under public information can fail when information is private and noisy [1]. A useful reputation system must therefore achieve two things at once. Observers should agree with one another, and their judgements should still distinguish cooperative behaviour from defection. We examined both properties in finite populations containing equal numbers of ALLC, ALLD and DISC players, in which each DISC observer keeps its own record of every individual and strategies are held fixed. We measure recognition by the reputational separation

$$\Delta g = \frac{g_{\mathrm{DISC}} + g_{\mathrm{ALLC}}}{2} - g_{\mathrm{ALLD}}, \tag{1}$$

where g_I is the probability that a DISC observer reads a type-I individual as good, and agreement by the mean disagreement $\bar{d}$, the probability that two observers read the same individual differently (Methods).

Without inertia, each norm fails in a characteristic way (Fig. 2a–d, top row). Under Scoring, unconditional cooperators are regarded as good and defectors as bad, but observers are divided about discriminators, whose justified refusals to help bad recipients are condemned. Shunning drives almost every reputation towards bad. Stern Judging sustains disagreement at chance level, and Simple Standing extends substantial trust to defectors. Full one-step inertia does not resolve these failures (Fig. 2a–d, middle row). Shunning provides the clearest example: observers now agree almost perfectly, but only because they regard almost everyone as bad ($g_I < 0.001$ for all three strategies). Agreement alone is therefore not the right target. What matters is agreement on judgements that still discriminate between behaviours.

Both requirements can be met, but only when inertia is allocated in a norm-specific direction (Fig. 2e–h). Using $\Delta g \geq 0.8$ and $\bar{d} \leq 0.1$ as operational criteria, the qualifying allocations with the smallest total depth are $(k_1, k_2) = (2,0)$ under Scoring, $(3,0)$ under Shunning, $(4,3)$ under Stern Judging and $(1,2)$ under Simple Standing. In replicated simulations, these allocations give separations of 0.88 to 0.92 with mean disagreement between 0.06 and 0.08, whereas full one-step inertia meets both criteria under none of the norms (Extended Data Fig. 1a–d). Their multi-step reputation matrices show that recognition and agreement across observers are recovered together (Fig. 2a–d, bottom row). The exact integers depend on the chosen criteria, but the direction of the qualifying allocations does not (Extended Data Fig. 1e–h). Direction can matter as much as amount: under Stern Judging, most allocations with more recovery than warning states invert recognition, so that defectors are regarded better than unconditional cooperators. We use these four allocations as representative multi-step architectures below. They were selected from reputation statistics alone, not by maximizing cooperation, invasion resistance or the size of any basin of attraction.

The selected directions follow the errors that each norm introduces into private assessments. Scoring and Shunning condemn discriminators who justifiably refuse to help bad recipients, so cooperative individuals accumulate unwarranted negative assessments, and these norms benefit from inertia on the deterioration path. Simple Standing approves of defection against bad recipients, so defectors can regain good standing, and this norm benefits from stricter recovery. Stern Judging makes the verdict on any action depend on the observer's private view of the recipient, which exposes both cooperators and defectors to erroneous assessments and calls for inertia in both directions. This generalization is suggested by the four norms studied here; it is not a general theorem.

Recognition concerns how different strategies are judged. A related question is whether a population of discriminators can sustain a consistent high-reputation state among themselves. In a homogeneous DISC population, full one-step inertia makes Scoring, Shunning and Stern Judging bistable (Extended Data Fig. 2a–d). Besides a high-reputation state, these norms sustain a stable state in which discriminators regard one another as bad (Scoring, Shunning) or as good only half of the time (Stern Judging). Which state the population reaches therefore depends on its reputational history. The representative multi-step architectures have a single, high-reputation state, so this history dependence disappears. Scans over the readout parameters and the depths show that both reputation levels and multistability are reorganized by the direction of inertia, at the level of readout as well as depth (Extended Data Fig. 3). Together, these results show that the relevant quantity is not memory depth in isolation, but the direction in which evidence is accumulated. Appropriately directed inertia turns noisy private information into reputations that discriminate between behaviours and are consistent across observers. Having established this, we next ask whether these reputational changes alter the evolution of cooperation.

Directional inertia reshapes strategy dynamics

We let ALLC, ALLD and DISC evolve according to the replicator dynamics, with payoffs computed from the private reputations that relax from initially good records at each strategy composition (Methods). Without inertia, instantaneous private updating

provides no common route to stable cooperation (Fig. 3, top row). Under Shunning and Stern Judging, every trajectory leads to defection. Under Scoring and Simple Standing, cooperative configurations exist, but in neither norm is a population of discriminators stable; cooperation is maintained in mixtures dominated by unconditional cooperators.

Full one-step inertia changes this picture qualitatively (Fig. 3, middle row). It makes a population of discriminators a strict local attractor under all four norms at the reference parameters, so highly cooperative states now exist across the norms. Their accessibility, however, remains strongly norm dependent. Averaged over initial strategy compositions sampled uniformly from the simplex, only 8% of compositions lead to cooperation under Shunning and 23% under Stern Judging. Inertia can thus create or stabilize cooperative states while the route to them still depends on how reputations form.

The selected multi-step architectures reorganize the strategy landscape further (Fig. 3, bottom row). Under Scoring, Shunning and Stern Judging, a population of discriminators remains a strict attractor and is reached from a larger part of the simplex: 88%, 71% and 71% of initial compositions, respectively. Under Simple Standing, the cooperative attractor is not a pure discriminator population but a stable mixture of about 80% discriminators and 20% unconditional cooperators, reached from 76% of initial compositions. In all four norms, the cooperative attractors achieve realized cooperation between 0.975 and 0.980, close to the maximum of $1-e_1 = 0.98$ set by implementation errors. The errors remain part of the model; the selected architectures sustain high cooperation despite them. Mean long-run cooperation, averaged over initial compositions, rises accordingly, from 0.61 without inertia to 0.87 under Scoring, from 0 to 0.69 under Shunning, from 0 to 0.70 under Stern Judging and from 0.54 to 0.74 under Simple Standing. The broader accessibility of cooperation accounts for much of this increase, but the cooperative states themselves also change: with inertia they are sustained by discriminators rather than by majorities of unconditional cooperators. Defection is never eliminated; a population of defectors remains a strict attractor in every case. These outcomes are also robust across contrasting initial reputations (Extended Data Fig. 2e). Starting from all-bad instead of all-good records leaves them unchanged under Scoring, Shunning and Simple Standing, and changes mean cooperation under Stern Judging only slightly (0.699 versus 0.706). One-step outcomes under Shunning and Stern Judging, by contrast, lose all cooperation when records start bad.

Whether an established discriminator population can resist rare unconditional mutants has a compact analytical answer. Let r_{R}^* be the probability that residents are regarded as good, and let $u_C(r)$ and $u_D(r)$ be the probabilities that intended cooperation and intended defection receive a positive assessment when a fraction r of recipients is regarded as good (Methods). A rare mutant does not change the resident assessment environment, so its reputation follows from the stationary response $\mathscr{R}(u)$ of the record, the probability of being read as good when assessments are independently positive with probability u. When an ALLD mutant is regarded worse, and an ALLC mutant better, than the residents, the discriminator population resists both mutants if and only if

$$R_L \equiv \frac{r_{\mathrm{R}}^*}{r_{\mathrm{R}}^* - \mathscr{R}\left[u_D(r_{\mathrm{R}}^*)\right]} < \frac{b}{c} < \frac{1-r_{\mathrm{R}}^*}{\mathscr{R}\left[u_C(r_{\mathrm{R}}^*)\right] - r_{\mathrm{R}}^*} \equiv R_U. \tag{2}$$

The lower bound requires defectors to lose enough help to offset the cooperation costs they save. The upper bound arises because ALLC also helps recipients that discriminators regard as bad, and can gain a reputational advantage from doing so. Without inertia, this window is empty under all four norms. Full one-step inertia opens it, with R_U between 5.6 and 6.3, and the selected architectures extend it to $R_U > 200$ under Scoring, Shunning and Stern Judging. Under Simple Standing, the selected architecture has $R_U \approx 4.0$, below the reference ratio $b/c = 5$. A discriminator population can therefore be invaded by ALLC, and the population settles in the highly cooperative ALLC–DISC mixture rather than losing cooperation.

Why do one-step and multi-step inertia produce such different evolutionary geometries? The answer lies in how the reputation of discriminators deteriorates as defectors become common (Extended Data Fig. 4). Along the edge of the simplex where only ALLD and DISC are present, one-step inertia under Shunning and Stern Judging sustains a high-reputation state of discriminators only above a critical frequency ($f_{\mathrm{DISC}} \approx 0.97$ and 0.77, respectively). Below it, this state disappears at a fold; the reputation of discriminators collapses, and with it their payoff advantage over defectors. The selected Shunning architecture moves this tipping point to much lower frequencies ($f_{\mathrm{DISC}} \approx 0.51$). The selected Stern Judging architecture removes the fold altogether and replaces it with a smooth threshold at which discriminators begin to outperform defectors ($f_{\mathrm{DISC}} \approx 0.28$). Under Scoring and Simple Standing, the change is not driven by folds; inertia smoothly shifts this threshold towards lower frequencies of discriminators. In all four norms, directional inertia therefore changes how the reputation of discriminators erodes as defectors spread, and thereby moves the population composition at which discriminators gain an evolutionary advantage. Folds describe the abrupt form this change takes under Shunning and Stern Judging; they are not the general mechanism.

The advantage of the selected architectures persists across benefit-to-cost ratios: they give the highest mean cooperation of the three settings under every norm for all $b/c \geq 1.1$ (Extended Data Fig. 5). Across implementation and assessment errors up to 0.45, they also broaden the region in which a discriminator population resists both unconditional mutants under Scoring, Shunning and Stern Judging, but not under Simple Standing (Extended Data Fig. 6). This error analysis concerns

local resistance only; mean cooperation was not scanned across error rates. Taken together, directional inertia reorganizes reputation-dependent selection across population compositions. It produces highly cooperative attractors that are accessible from a larger part of the strategy simplex.

Cooperation depends on how inertia is allocated, not simply on total depth

The four representative architectures differ in their total depth $K = k_1 + k_2$. Is their success merely a consequence of having more intermediate states? To separate amount from direction, we fixed K and compared every allocation between warning and recovery states (Fig. 4). At fixed total depth, mean long-run cooperation spans a wide range. With eight intermediate states under Scoring, it ranges from almost zero, when all are recovery states, to 0.875, when all are warning states. Allocating every state to recovery yields no cooperation under Scoring, Shunning and Stern Judging at any depth. Total depth alone therefore does not predict the evolutionary outcome.

The most cooperative allocation at each depth follows the same norm-specific directions that emerged from recognition. Scoring and Shunning favour pure tolerance. Stern Judging favours tolerance at small depths and substantial inertia in both directions from $K = 3$ onwards. Simple Standing favours a shallow warning path combined with a deeper recovery path from $K = 2$ onwards. Notably, all four architectures chosen from reputation statistics alone lie on this cooperation envelope at their respective depths. This links the quality of private reputations to evolutionary performance without circular parameter selection. The full (k_1, k_2) plane confirms that these patterns persist beyond the one-dimensional comparison at fixed depth (Extended Data Fig. 7).

More inertia is not automatically better. Under Scoring, the best achievable cooperation peaks at $K = 5$ (0.888) and declines slightly as more warning states are added (0.875 at $K = 8$). This is consistent with longer warning paths also delaying the condemnation of defectors. Under Stern Judging, the best achievable cooperation does not increase from $K = 7$ to $K = 8$. Temporal integration of evidence must be allocated appropriately rather than maximized indiscriminately.

Directionality is present at the level of readout as well as depth (Extended Data Fig. 8). Whereas Fig. 4 varies how many intermediate states exist, the readout parameters vary how strongly those states affect current treatment. At one-step depths, tolerance-only readout, $(\alpha, \beta) = (1, 0)$, makes a discriminator population a strict local attractor under all four norms, whereas strictness-only readout, $(0, 1)$, does so under none. The detailed response is norm dependent. Under Scoring, Shunning and Stern Judging, mean cooperation is highest at tolerance-only readout and vanishes at strictness-only readout, whereas under Simple Standing cooperation remains high across the entire readout plane. Inertia is thus a directional architecture rather than a scalar amount of memory, and the beneficial allocation follows the errors that the social norm introduces into private assessments.

Reputation readout balances tolerance and opportunistic exploitation

Tolerance of negative evidence is a common feature of the successful architectures. This raises a natural question. If isolated negative evidence is forgiven, can a strategy deliberately generate such evidence occasionally and exploit the same protection? The mutants considered so far defect either always or never. We therefore introduce an opportunistic discriminator (O-DISC), which behaves like DISC except that it withholds intended help from a recipient it regards as good with probability δ, independently in every interaction. This family interpolates continuously between DISC ($\delta = 0$) and ALLD ($\delta = 1$). In a resident population of discriminators, a rare O-DISC mutant has the normalized invasion payoff

$$\Lambda(\delta) = \frac{\Pi_X - \Pi_R}{c(1-e_1)} = \delta r_R^* - \frac{b}{c}\left[r_R^* - r_X^*(\delta)\right], \tag{3}$$

where $r_X^*(\delta)$ is the probability that the mutant is regarded as good. The first term is the cooperation cost the mutant saves. The second is the loss of future benefits caused by the reputational damage that its omissions produce. The mutant can invade if $\Lambda > 0$.

Resistance to ALLD does not imply resistance to partial opportunism (Fig. 5a). Under Stern Judging with the selected architecture $(4, 3)$ and full readout, residents are regarded as good with probability $r_R^* > 0.9999$, and ALLD cannot invade. Nonetheless, every O-DISC mutant with $0 < \delta < 0.365$ can. Full one-step inertia leaves a much narrower window ($0 < \delta < 0.008$). Similar windows arise in two representative alternative mechanisms that also buffer isolated negative evidence: aggregation of two sampled assessments [2] and a bounded quantitative score [3]. Both exclude ALLD but admit intermediate opportunists. These comparisons are not performance benchmarks; their parameters were not optimized and the mechanisms

are not matched in memory. They indicate that the vulnerability is not unique to the inertial state architecture and point to a broader trade-off in tolerant reputation processing.

The origin of the window lies in the response to weak opportunism. Differentiating Eq. (3) at $\delta = 0$ gives

$$\Lambda'(0) = r_{\mathrm{R}}^{*}\left[1 - \frac{b}{c}\,\Delta u(r_{\mathrm{R}}^{*})\,\partial_u \mathscr{R}(u_{\mathrm{R}}^{*};\alpha,\beta,k_1,k_2)\right], \tag{4}$$

where $\Delta u(r) = u_C(r) - u_D(r)$ is the assessment contrast between intended cooperation and defection, and u_{R}^{*} is the probability that residents receive a positive assessment. If the reputation response is locally too flat near the resident state, a small amount of opportunism saves more cooperation cost than it loses in future benefits. Full tolerance readout produces exactly such a flat response. For any record with a warning path ($k_1 \geq 1$), $\partial_u \mathscr{R} \to 1-\alpha$ as $u \to 1$, independently of the other parameters. With $\alpha = 1$, the only remaining penalty comes from the chance of completing the warning path, which is small when that path is long and residents are almost always assessed positively.

Lowering α makes warning states socially consequential before a record reaches the bad state. It increases the marginal reputational penalty for occasional defection without changing how evidence is accumulated. For the Stern Judging architecture, reducing α from 1 to 0.8 removes the invasion window entirely, while residents remain highly cooperative ($r_{\mathrm{R}}^{*} = 0.987$; Fig. 5a). More generally, residents exclude every O-DISC mutant in an interval $\alpha_L < \alpha \leq \alpha_U$ (Fig. 5b), whose boundaries follow from the resident equilibrium $F(r,\alpha) = 0$, with $F(r,\alpha) = \mathscr{R}(u_{\mathrm{R}}(r);\alpha,\beta,k_1,k_2) - r$. The lower boundary is a fold of the high-reputation resident state,

$$F(r_L,\alpha_L) = 0, \qquad \partial_r F(r_L,\alpha_L) = 0. \tag{5}$$

Below α_L, warning states are trusted too little to sustain the high-reputation state; in all cases we computed, the residents then fall to a low-reputation state that opportunists can invade. The upper boundary is where the largest invasion payoff of the O-DISC family becomes neutral, $F(r_U,\alpha_U) = 0$ and $\max_{\delta\in(0,1]}\Lambda(\delta;\alpha_U) = 0$. In all cases we computed, security is first lost to infinitesimally weak opportunism, so that this condition reduces to $\Lambda'(0;\alpha_U) = 0$, or

$$\frac{b}{c}\,\Delta u(r_U)\,\partial_u \mathscr{R}(u_{\mathrm{R}}(r_U);\alpha_U,\beta,k_1,k_2) = 1. \tag{6}$$

For the Stern Judging architecture, $\alpha_L \approx 0.567$ and $\alpha_U \approx 0.804$. Both boundaries depend on the norm, the depths, β and the error rates through the full resident map, so neither is a universal threshold. Where no fold occurs within the readout range, the secure interval extends down to $\alpha = 0$.

Protective readouts exist under all four norms, although their locations differ (Extended Data Fig. 9). At the common illustrative readout $\alpha = 0.8$ and $b/c = 5$, all four selected architectures are secure against the tested O-DISC family, while their residents remain highly cooperative (r_{R}^{*} between 0.978 and 0.990). Under one-step inertia, the protective intervals of Shunning and Stern Judging lie close to full readout, and at $\alpha = 0.8$ these resident populations have already lost their high-reputation state. Because resident reputations do not depend on b/c, the fold boundary α_L is unaffected by the benefit-to-cost ratio. The upper boundary α_U instead rises with b/c, since larger benefits let a smaller reputational loss deter opportunism. At low benefit-to-cost ratios, the interval can therefore close (Shunning at $b/c = 2$). These results concern the specified family of independent, identically distributed omissions. They do not establish resistance to other forms of exploitation, such as persistent or temporally correlated opportunism, nor to ALLC; under Simple Standing, the selected architecture is secure against opportunists but can be invaded by unconditional cooperators. Inertia controls how behavioural evidence is accumulated over time, whereas readout controls how much incomplete negative evidence already carries social consequences. Because these two controls are separable, a reputation system can tolerate accidental negative evidence without making deliberate occasional defection costless.

Discussion

To be completed following the author's outline.

Methods

Donation game, strategies and private assessment

We consider a well-mixed population in which a donor can pay a cost c to provide a benefit $b > c$ to a recipient. Defection provides no benefit and incurs no cost. The resident strategy set is $\mathscr{S} = \{\mathrm{ALLC}, \mathrm{ALLD}, \mathrm{DISC}\}$. ALLC always intends to

cooperate, ALLD always intends to defect, and DISC intends to cooperate if its private image of the recipient is good and to defect otherwise. In the deterministic analyses, f_I denotes the frequency of strategy I, with $\sum_I f_I = 1$. Donor and recipient roles are sampled without self-interaction.

Reputations are private. A DISC donor chooses its action using its own image of the recipient, whereas an observer assesses the realized action using the observer's image of that recipient. The four second-order norms share positive assessment of cooperation with a good recipient and negative assessment of defection against a good recipient. Their prescriptions for a recipient read as bad are parameterized by p and q, the probabilities of positive assessment after cooperation and defection, respectively:

$$\phi(C,G) = 1, \qquad \phi(D,G) = 0, \qquad \phi(C,B) = p, \qquad \phi(D,B) = q. \tag{7}$$

Scoring, Shunning, Stern Judging and Simple Standing correspond to $(p,q) = (1,0),(0,0),(0,1)$ and $(1,1)$, respectively.

Intended cooperation fails and becomes defection with probability e_1; intended defection is implemented without error. After applying the norm, the observer reverses the prescribed assessment with probability e_2. These errors occur independently. Let $P^+_{A|R}$ be the probability of a recorded positive assessment conditional on intended action $A \in \{C,D\}$ and the observer's image $R \in \{G,B\}$ of the recipient. Defining $\varepsilon = (1-e_1)(1-e_2) + e_1 e_2$, we obtain

$$\begin{aligned} P^+_{C|G} &= \varepsilon, & P^+_{D|G} &= e_2, \\ P^+_{C|B} &= p(\varepsilon - e_2) + q(1-\varepsilon-e_2) + e_2, & P^+_{D|B} &= q(1-2e_2) + e_2. \end{aligned} \tag{8}$$

The action subscript in Eq. (8) refers to the intended action; implementation error has already been averaged into these probabilities. Unless varied explicitly, $e_1 = e_2 = 0.02$, $c = 1$ and $b/c = 5$.

Inertial records and their binary readout

For every ordered observer–target pair (i,j), the observer retains a latent record L_{ij} in the state space

$$\mathscr{L} = \{G, W_1, \ldots, W_{k_1}, B, S_1, \ldots, S_{k_2}\}, \qquad k_1, k_2 \in \mathbb{Z}_{\geq 0}. \tag{9}$$

The warning states accumulate evidence against established good standing. A negative assessment advances $G \to W_1 \to \cdots \to W_{k_1} \to B$, whereas a positive assessment sends every W_m directly to G. The recovery states accumulate evidence in favor of restoring good standing. A positive assessment advances $B \to S_1 \to \cdots \to S_{k_2} \to G$, whereas a negative assessment sends every S_n directly to B. Positive assessments leave G unchanged and negative assessments leave B unchanged. An absent path is traversed in one update: $G \to B$ after a negative assessment when $k_1 = 0$, and $B \to G$ after a positive assessment when $k_2 = 0$. Thus, reversal requires $k_1 + 1$ consecutive negative assessments or $k_2 + 1$ consecutive positive assessments. Evidence supporting the established judgment resets the entire opposing path.

Actions and assessments consult a binary image sampled from the latent record. Its good-readout probability is

$$\sigma(\ell) = \begin{cases} 1, & \ell = G, \\ \alpha^m, & \ell = W_m, \\ 0, & \ell = B, \\ 1-\beta^n, & \ell = S_n, \end{cases} \qquad 0 \leq \alpha, \beta \leq 1. \tag{10}$$

Every consultation resamples this binary image independently; consultation does not alter L_{ij}. Increasing α increases tolerance of negative evidence, whereas increasing β increases strictness about rehabilitation. With $\alpha = \beta = 1$, warnings are read as good and recovery states as bad until a path is completed. The one-step model has $k_1 = k_2 = 1$. At these depths, $(\alpha,\beta) = (0,0)$ reproduces instantaneous binary updating, $(1,0)$ gives tolerance alone, $(0,1)$ gives strictness alone, and $(1,1)$ gives full one-step inertia. The depth pair $(0,0)$ is the binary baseline for the depth scans.

Stationary assessment-to-reputation map

Let u be the probability of a positive recorded assessment in a fixed assessment environment. Write T^+ and T^- for the transition matrices corresponding to positive and negative assessments. The latent chain has transition matrix $T(u) = uT^+ + (1-u)T^-$.

Its stationary distribution satisfies $\boldsymbol{\pi} = \boldsymbol{\pi} T(u)$ and $\sum_\ell \pi_\ell = 1$. Averaging Eq. (10) over this distribution gives the stationary probability of being read as good:

$$\mathscr{R}(u;\alpha,\beta,k_1,k_2) = \frac{u^{k_2+1}\left[1+\sum_{m=1}^{k_1}\{\alpha(1-u)\}^m\right] + (1-u)^{k_1+1}\sum_{n=1}^{k_2} u^n(1-\beta^n)}{(1-u)^{k_1} + u^{k_2} - (1-u)^{k_1}u^{k_2}}. \tag{11}$$

Empty sums are zero; the endpoint values are $\mathscr{R}(0) = 0$ and $\mathscr{R}(1) = 1$. For one-step inertia,

$$\mathscr{R}(u) = \frac{u^2[1+\alpha(1-u)] + (1-\beta)u(1-u)^2}{1-u+u^2}. \tag{12}$$

Setting $\alpha = \beta = 0$ gives $\mathscr{R}(u) = u$. The mapping describes stationary readout probabilities under the specified assessment environment; it does not identify the transient binary readout with a Markov process on two states.

Private-reputation self-consistency and branch selection

For fixed strategy frequencies, r_I denotes the probability that a type-I target is read as good. We use a mean-field closure in which the donor's and observer's private images of a recipient are conditionally independent given the recipient's strategy. Accordingly,

$$\begin{aligned} \bar{r} &= \sum_I f_I r_I, & h_{GG} &= \sum_I f_I r_I^2, \\ h_{BB} &= \sum_I f_I (1-r_I)^2, & h_{GB} &= \sum_I f_I r_I (1-r_I). \end{aligned} \tag{13}$$

Here h_{GB} is the probability of each ordered disagreement, so the total disagreement probability is $2h_{GB}$. The positive-assessment probabilities generated by the three strategies are

$$\begin{aligned} \Phi_{\mathrm{ALLC}} &= \bar{r}P^+_{C|G} + (1-\bar{r})P^+_{C|B}, \\ \Phi_{\mathrm{ALLD}} &= \bar{r}P^+_{D|G} + (1-\bar{r})P^+_{D|B}, \\ \Phi_{\mathrm{DISC}} &= h_{GG}P^+_{C|G} + h_{GB}(P^+_{C|B} + P^+_{D|G}) + h_{BB}P^+_{D|B}. \end{aligned} \tag{14}$$

Stationary reputations satisfy $r_I^* = \mathscr{R}(\Phi_I^*)$. The deterministic reputation calculations select a solution from initially good records. For continuous equilibration, we use the auxiliary relaxation

$$\frac{\mathrm{d}r_I}{\mathrm{d}\tau} = \mathscr{R}(\Phi_I) - r_I, \qquad r_I(0) = 1. \tag{15}$$

This equation has the required stationary self-consistency solutions as its fixed points. Its auxiliary time τ is distinct from the microscopic event clock. The corresponding asynchronous latent-state mean-field system is $\dot{\boldsymbol{\pi}}_I = \boldsymbol{\pi}_I[T(\Phi_I) - \mathbf{I}]$, with $r_I = \boldsymbol{\pi}_I \boldsymbol{\sigma}$; its stationary equations give the same mapping within this closure. Equality of stationary equations alone does not establish equality of transient dynamics or attraction basins.

When a fraction r of recipients is regarded as good and observers' images are conditionally independent, intended cooperation and intended defection receive positive assessments with probabilities

$$\begin{aligned} u_C(r) &= rP^+_{C|G} + (1-r)P^+_{C|B}, & u_D(r) &= rP^+_{D|G} + (1-r)P^+_{D|B}, \\ u_{\mathrm{R}}(r) &= r\,u_C(r) + (1-r)\,u_D(r), & \Delta u(r) &= u_C(r) - u_D(r). \end{aligned} \tag{16}$$

In a homogeneous DISC population, $u_{\mathrm{R}}(r)$ equals Φ_{DISC} in Eq. (14). We locate the roots of $F(r) = \mathscr{R}[u_{\mathrm{R}}(r)] - r$ and classify a non-degenerate root as stable or unstable in the scalar relaxation according to the sign of $F'(r)$. The all-good-selected scalar solution is the highest stable root. Extended Data Fig. 2 reports the stationary roots and the stability of this reduced flow. Root finding partitions $[0,1]$ at the extrema of F before applying bracketed scalar root finding, which permits detection of roots near folds. The drift and depth scans use 2,001-point search grids; the one-step readout plane uses a 161×161 grid and a 401-point root-search grid, with finer-grid checks. These classifications refer to the scalar closure; they do not establish the stability of the full latent-state system.

Finite-population simulations and recognition statistics

The quantitative individual-based reputation simulations contain $N = 150$ individuals, with 50 individuals of each strategy. The illustrative image matrices use separately saved realizations with $N = 90$, comprising 30 individuals of each strategy. Frequencies remain fixed throughout each run. We store the latent records held by the $m = n_{\mathrm{DISC}}$ DISC observers for all N targets, including self-records. Each event draws a donor and a distinct recipient uniformly from the population, then draws an observer independently and uniformly from the DISC individuals. The observer may coincide with the donor or recipient. The donor selects its intended action, implementation error determines the realized action, and the observer applies the norm using a separately sampled private readout of the recipient. Assessment error determines the recorded sign, which updates the observer's latent record of the donor. This simulation retains target-specific variation and correlations between private records rather than imposing the conditional-independence closure.

One sweep consists of mN events, corresponding to one update per retained record on average. All records start in G. The recognition depth grid varies k_1, k_2 independently over $0, \ldots, 6$ at $\alpha = \beta = 1$. The depth map in Fig. 2 uses one run per condition; the selection and sensitivity analysis in Extended Data Fig. 1 use 30. The standard protocol runs for 1,000 sweeps and samples every five sweeps from sweep 800 through sweep 1,000, yielding 41 measurements per run. Two slowly relaxing Stern Judging conditions use longer horizons: $(k_1, k_2) = (4, 5)$ uses 3,000 sweeps with 2,400 discarded, and $(3, 5)$ uses 50,000 sweeps with 40,000 discarded. Their sampling interval and replicate count are unchanged. Condition-specific seeds are generated from base seed 20260918, norm, depths and replicate index. Matrix panels are separately saved illustrative snapshots; the quantitative depth statistics use time-averaged replicate estimates.

Let $s_{ij} = \sigma(L_{ij})$. For target j, its mean goodness and disagreement between distinct DISC observers are

$$g_j = \frac{1}{m}\sum_i s_{ij}, \qquad d_j = \frac{1}{\binom{m}{2}}\sum_{i<\ell}\left[s_{ij}(1-s_{\ell j}) + (1-s_{ij})s_{\ell j}\right]. \tag{17}$$

These estimators average over stochastic readout exactly. We average over targets of strategy I to obtain g_I and d_I, then over sampling times within each replicate. Reputational separation and target-weighted mean disagreement are

$$\Delta g = \frac{g_{\mathrm{DISC}} + g_{\mathrm{ALLC}}}{2} - g_{\mathrm{ALLD}}, \qquad \bar{d} = \sum_I \frac{n_I}{N} d_I. \tag{18}$$

We report arithmetic means across independent replicate means. Where uncertainty is evaluated, pointwise 95% percentile intervals use 2,000 bootstrap resamples of these replicate means. We assess stationarity by comparing the early and late halves of the sampling window; a drift interval contained within $[-0.01, 0.01]$ is a practical small-drift diagnostic, rather than a proof of mixing.

The cutoffs $\Delta g \geq 0.8$ and $\bar{d} \leq 0.1$ are operational selection criteria. Among qualifying architectures, we minimize total depth $K = k_1 + k_2$, then break ties by greater separation and lower disagreement. This gives $(2,0), (3,0), (4,3)$ and $(1,2)$ for Scoring, Shunning, Stern Judging and Simple Standing. These pairs are selected from reputation formation, independently of evolutionary performance. Extended Data Fig. 1 varies the separation cutoff from 0.70 to 0.90 in steps of 0.05 and the disagreement cutoff from 0.050 to 0.150 in steps of 0.025.

Payoffs, strategy evolution and cooperation

Reputation formation is assumed to be fast relative to strategy evolution. For each strategy composition, we compute stationary reputations before evaluating expected payoffs:

$$\begin{aligned}
\Pi_{\mathrm{ALLC}} &= (1-e_1)\{b(f_{\mathrm{ALLC}} + f_{\mathrm{DISC}} r^*_{\mathrm{ALLC}}) - c\},\\
\Pi_{\mathrm{ALLD}} &= (1-e_1) b(f_{\mathrm{ALLC}} + f_{\mathrm{DISC}} r^*_{\mathrm{ALLD}}),\\
\Pi_{\mathrm{DISC}} &= (1-e_1)\{b(f_{\mathrm{ALLC}} + f_{\mathrm{DISC}} r^*_{\mathrm{DISC}}) - c\bar{r}^*\}.
\end{aligned} \tag{19}$$

Strategy frequencies obey

$$\frac{\mathrm{d}f_I}{\mathrm{d}t} = f_I(\Pi_I - \bar{\Pi}), \qquad \bar{\Pi} = \sum_J f_J \Pi_J. \tag{20}$$

The intended and realized cooperation rates are, respectively,

$$\rho^{\mathrm{int}} = f_{\mathrm{ALLC}} + f_{\mathrm{DISC}}\bar{r}^*, \qquad \rho = (1-e_1)\rho^{\mathrm{int}}. \tag{21}$$

Thus, an intended cooperation rate approaching one corresponds to realized cooperation approaching $1-e_1$. All cooperation averages over initial compositions report realized cooperation; the opportunism analyses report the resident reputation r_{R}^*, which equals the intended cooperation of a DISC population.

For global comparisons we integrate the replicator dynamics from M initial strategy compositions sampled uniformly with respect to area on the three-strategy simplex. Every trajectory converges either to the set without cooperation or to an attractor with positive cooperation, and the two classes are separated by a wide gap in final realized cooperation; we therefore classify each initial condition by the attractor it reaches. If $\boldsymbol{f}_{\mathrm{end}}^{(m)}$ is the terminal composition from initial condition m and $\mathscr{C}$ the set of initial conditions reaching a cooperative attractor, we report the mean long-run cooperation, the size of the cooperative basin and the cooperation level within it,

$$\bar{\rho}=\frac{1}{M}\sum_{m=1}^{M}\rho\left(\boldsymbol{f}_{\mathrm{end}}^{(m)}\right),\qquad \omega_C=\frac{|\mathscr{C}|}{M},\qquad \bar{\rho}_C=\frac{1}{|\mathscr{C}|}\sum_{m\in\mathscr{C}}\rho\left(\boldsymbol{f}_{\mathrm{end}}^{(m)}\right), \tag{22}$$

so that $\bar{\rho}\approx\omega_C\bar{\rho}_C$. We use common initial compositions when comparing parameters. For the total-depth comparison, we enumerate all allocations $k_1+k_2=K$ and define the upper envelope as the largest observed mean long-run cooperation at each K. This is a maximum over the scanned allocations at fixed depth, rather than a global optimum over unrestricted mechanisms.

Numerical strategy calculations

The supplied simplex panels evaluate the stationary self-consistency map by synchronous iteration of the two private-image moments $\bar{r}$ and h_{GG}, initialized at one. The stopping criterion is the sum of the absolute changes in these moments below 10^{-10}, with at most 1,000 iterations. The resulting payoff field is evaluated on a rectangular grid clipped to the simplex, with 84×74 grid positions before clipping, to draw streamlines. The plotting routine retains terminal iterates when the stationary-map iteration does not converge. Such points occur under Stern Judging and Simple Standing with inertia; their payoff field and associated basin geometry require a converged reputation calculation before being interpreted as stationary dynamics. Vertex stability uses the two rare-strategy growth rates; the displayed filled or open marker at an edge root refers to the eigenvalue tangent to that invariant edge. The transverse invasion eigenvalue is calculated separately. These edge classifications do not establish stability against a strategy absent from that edge. Synchronous iteration is a numerical stationary-map solver, and its convergence is distinct from the stability of Eq. (15).

The parameter scans integrate the continuous reputation relaxation by fourth-order Runge–Kutta with step 0.1, using residual tolerances of 10^{-9} or 10^{-10}. Slowly relaxing states are continued with step 0.2 for up to 80,000 additional steps. The equilibrated strategy vector field is tabulated on a triangular mesh and interpolated piecewise linearly for midpoint integration of strategy trajectories. Terminal reputation and cooperation values are recomputed without interpolation. The total-depth scan uses a mesh resolution of 23 and 512 common scrambled Sobol initial compositions, midpoint step 0.1, and up to eight blocks of 4,000 steps. The Scoring binary baseline uses its analytical stationary reputations and midpoint step 0.05. Its mean-cooperation stopping rule requires a change below 2×10^{-5} for two successive block comparisons. Three one-step depth points, under Scoring, Shunning and Simple Standing, did not pass this observable-stability gate within the saved horizon; they are finite-time terminal estimates, not confirmed stationary averages.

The benefit-to-cost scan in Extended Data Fig. 5 uses eight independent scrambled Sobol sets of 512 initial compositions, with seeds 2026–2033 and common compositions within each set. Means and standard errors are calculated across the eight set-level means. This uncertainty concerns the distribution of initial strategy compositions; the dynamical model itself is deterministic. The saved calculations use triangular mesh resolutions of 41 for Scoring and 25 for the other norms, midpoint steps 0.05 and 0.1, respectively, and up to 64 integration blocks. Their stopping gate checks changes in both mean and individual intended cooperation and the 99th percentile of the exact terminal strategy residual for successive blocks. Local invasion boundaries are deterministic and do not depend on the initial-condition samples.

Local resistance to ALLC and ALLD

For a pure-DISC population on the selected resident branch, write r_{R}^* for the resident reputation. A rare unconditional mutant does not change the resident environment and receives the reputations

$$\begin{aligned} r_C^* &= \mathscr{R}[u_C(r_{\mathrm{R}}^*)]\,,\\ r_D^* &= \mathscr{R}[u_D(r_{\mathrm{R}}^*)]\,. \end{aligned} \tag{23}$$

Their invasion growth rates relative to the resident are

$$\begin{aligned}\lambda_D &= (1-e_1)\{br_D^* - (b-c)r_{\mathrm{R}}^*\},\\ \lambda_C &= (1-e_1)\{br_C^* - c - (b-c)r_{\mathrm{R}}^*\}.\end{aligned} \tag{24}$$

When $r_{\mathrm{R}}^* > r_D^*$ and $r_C^* > r_{\mathrm{R}}^*$, strict resistance to both mutants requires

$$R_L = \frac{r_{\mathrm{R}}^*}{r_{\mathrm{R}}^* - r_D^*} < \frac{b}{c} < R_U = \frac{1-r_{\mathrm{R}}^*}{r_C^* - r_{\mathrm{R}}^*}. \tag{25}$$

If $r_{\mathrm{R}}^* \le r_D^*$, no finite benefit-to-cost ratio excludes ALLD. If $r_C^* \le r_{\mathrm{R}}^*$, ALLC imposes no finite upper threshold. We classify local outcomes directly from the signs of λ_D and λ_C, including cases without a finite threshold. A zero growth rate is a marginal boundary. These criteria refer to the two tested unconditional mutants and are separate from cooperative attraction basins.

Independent opportunistic under-cooperation and benchmark filters

An opportunistic discriminator, denoted O-DISC, follows DISC except that an intended cooperation is replaced by intended defection independently with probability δ. This deliberate choice occurs before the implementation error. In the pure-DISC resident environment, define

$$u_{\mathrm{R}}^* = u_{\mathrm{R}}(r_{\mathrm{R}}^*), \qquad \Delta u^* = \Delta u(r_{\mathrm{R}}^*), \tag{26}$$

with the functions of Eq. (16). Its positive-assessment probability and stationary reputation are

$$u_{\mathrm{X}}(\delta) = u_{\mathrm{R}}^* - \delta r_{\mathrm{R}}^* \Delta u^*, \qquad r_{\mathrm{X}}^*(\delta) = \mathscr{F}[u_{\mathrm{X}}(\delta)], \tag{27}$$

where $\mathscr{F}$ is the reputation filter used by the resident population. The payoff difference gives Eq. (3). The endpoints are a neutral DISC copy at $\delta = 0$ and ALLD at $\delta = 1$. Equation (4) tests sufficiently small departures from DISC; it does not alone exclude an interior interval of invading mutants.

For inertia, $\mathscr{F} = \mathscr{R}$. For the aggregation benchmark, based on Michel-Mata et al. [2], the filter assigns a good image when at least Q of M independently sampled assessments are positive, giving

$$\mathscr{F}_A(u) = \sum_{j=Q}^{M} \binom{M}{j} u^j (1-u)^{M-j}. \tag{28}$$

The displayed setting is $M = 2, Q = 1$, so $\mathscr{F}_A(u) = 2u - u^2$. For the quantitative benchmark, based on Schmid et al. [3], the filter uses an integer score $s \in \{-s_{\max}, \ldots, s_{\max}\}$, incremented by one after a positive assessment and decremented by one after a negative assessment, with reflection at the endpoints. Scores at least s_0 are read as good. With $t = u/(1-u)$,

$$\mathscr{F}_Q(u) = \frac{\sum_{j=s_{\max}+s_0}^{2s_{\max}} t^j}{\sum_{j=0}^{2s_{\max}} t^j}, \qquad 0 < u < 1. \tag{29}$$

The displayed setting is $s_{\max} = 5$, $s_0 = 0$, and endpoint values are obtained by their limits. Each filter is evaluated at its own all-good-selected self-consistent resident reputation under the same norm and errors. The comparisons concern these specified settings, rather than an optimization over aggregation or quantitative parameters.

The displayed invasion curves locate stationary resident roots on a 4,001-point grid, refine sign changes by bracketed root finding, and retain the highest stable root. The exploitation grid combines 4,001 uniformly spaced points on $[0,1]$, 1,001 points on $[0, 0.025]$, and 100 logarithmically spaced positive values from 10^{-9} to 10^{-3}; nonzero invasion crossings are refined by root finding. Aggregation and quantitative resident solutions are obtained from $r(0) = 1$ by damped stationary-map iteration, $r_{n+1} = 0.8 r_n + 0.2\mathscr{F}[u_{\mathrm{R}}(r_n)]$, until the step is below 2×10^{-14}.

Readout security and resident branch boundaries

For every readout setting, we recompute the resident reputation and its assessment environment. Define the reputation loss per unit of saved cooperation as

$$\begin{aligned} g_{\alpha,\beta}(\delta) &= \frac{r_{\mathrm{R}}^* - \mathscr{R}(u_{\mathrm{R}}^* - \delta r_{\mathrm{R}}^* \Delta u^*; \alpha, \beta, k_1, k_2)}{\delta r_{\mathrm{R}}^*}, \qquad \delta > 0,\\ g_{\alpha,\beta}(0) &= \Delta u^* \, \partial_u \mathscr{R}(u_{\mathrm{R}}^*). \end{aligned} \tag{30}$$

The derivative is with respect to the positive-assessment probability. Exclusion of the entire independently under-cooperating family is equivalent to

$$\frac{b}{c}\min_{0\le\delta\le1} g_{\alpha,\beta}(\delta) \ge 1. \tag{31}$$

Equality permits a marginal mutant; strict inequality gives strict exclusion. The minimum includes the analytical limit at zero, the endpoint at one and all admissible interior stationary points of the rational loss function. For the security planes, these stationary points are obtained from the real roots of its derivative numerator and checked within the physical interval. The numerical security tolerance is 10^{-10}.

For the high-reputation branch, the upper readout boundary α_U is defined by Eq. (31). A lower boundary α_L, when present, is the saddle-node at which that branch disappears. It is located from

$$F(r,\alpha)=0, \qquad \partial_r F(r,\alpha)=0, \quad F(r,\alpha)=\mathscr{R}[u_{\mathrm{R}}(r);\alpha,\beta,k_1,k_2]-r, \tag{32}$$

with the norm, errors, depths and β fixed. This fold is distinct from an invasion boundary and is excluded when a strictly stable high-reputation resident is required. We retain only physical resident roots with residual below 5×10^{-13} and reduced-flow slope below -10^{-9} in the refined security analysis; fold solutions are checked separately. The intended resident cooperation shown in these panels is r^*_{R}. Security statements concern O-DISC with independent under-cooperation over $0<\delta\le1$, together with the separately stated ALLC and ALLD checks; they do not constitute an unrestricted evolutionary-stability claim.

References

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Figure asset `figures/Fig1.pdf` is provided in the companion source archive.

Figure 1. Directional inertia separates the accumulation of behavioural evidence from the interpretation of incomplete evidence. a, Instantaneous updating: a single assessment replaces the observer's binary image of a donor. A negative assessment (pink arrow) turns a good image (G) bad (B), and a positive assessment (blue arrow) turns a bad image good. **b**, Inertial updating: negative assessments advance a record in good standing through k_1 warning states, $W_1, \ldots, W_{k_1}$, before it becomes B (deterioration path), and positive assessments advance a record in bad standing through k_2 recovery states, $S_1, \ldots, S_{k_2}$, before it becomes G (recovery path). Evidence consistent with the established judgement returns any warning state to G and any recovery state to B, and leaves G and B unchanged. Losing good standing therefore requires $k_1 + 1$, and regaining it $k_2 + 1$, consecutive assessments of the same individual by the same observer. When a record is consulted for an action or an assessment, warning state W_m is read as good with probability α^m and recovery state S_n is read as bad with probability β^n; labels show the readout of the last state on each path. The depths k_1, k_2 control when reputations reverse, whereas α, β control how incomplete evidence affects current treatment. Instantaneous updating corresponds to $k_1 = k_2 = 0$ or $\alpha = \beta = 0$.

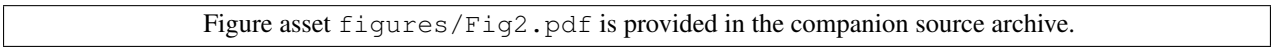


Figure 2. The direction in which inertia is allocated determines whether private observers both agree and recognize defectors. Finite populations with equal numbers of ALLC, ALLD and DISC players and fixed strategies; private records start good; $e_1 = e_2 = 0.02$. **a**–**d**, Private images held by DISC observers (rows) of DISC, ALLC and ALLD individuals (column blocks) under Scoring (**a**), Shunning (**b**), Stern Judging (**c**) and Simple Standing (**d**); blue, good; white, bad. Rows of blocks show no inertia (instantaneous updating), full one-step inertia ($k_1 = k_2 = 1$, $\alpha = \beta = 1$) and the selected multi-step architectures indicated above the blocks ($\alpha = \beta = 1$). Each block is one realization after 1,000 sweeps with $N = 90$. Without inertia and with one-step inertia, each norm fails differently; the selected architectures restore both recognition and agreement. **e**–**h**, Reputation quality across tolerance depth k_1 and strictness depth k_2 ($\alpha = \beta = 1$; $N = 150$; one simulation per pair, averaged over sweeps 800–1,000). Colour shows the separation Δg (Eq. (1)); circle area shows the mean disagreement $\bar{d}$, the probability that two DISC observers read the same individual differently. Black and red outlines mark no inertia, $(0,0)$, and one-step inertia, $(1,1)$. Stars mark the allocation with the smallest total depth satisfying $\Delta g \geq 0.8$ and $\bar{d} \leq 0.1$. These architectures are identified from reputation statistics only, not from evolutionary performance; replicated estimates and the dependence on the criteria are shown in Extended Data Fig. 1.

Figure asset `figures/Fig3.pdf` is provided in the companion source archive.

Figure 3. Directional inertia reshapes the evolutionary dynamics of cooperation. Replicator dynamics of ALLC, ALLD and DISC under Scoring (**a**), Shunning (**b**), Stern Judging (**c**) and Simple Standing (**d**). Rows show no inertia (top), full one-step inertia ($k_1 = k_2 = 1$, $\alpha = \beta = 1$; middle) and the selected multi-step architectures ($\alpha = \beta = 1$; $(k_1, k_2) = (2,0)$, $(3,0)$, $(4,3)$ and $(1,2)$, respectively; bottom). Vertices correspond to monomorphic populations. Without inertia, a population of discriminators is never stable. One-step inertia makes it a strict local attractor under all four norms, but cooperation remains reachable from only a small part of the simplex under Shunning and Stern Judging. The selected architectures sustain highly cooperative attractors reached from a larger part of the simplex: a pure discriminator population under Scoring, Shunning and Stern Judging, and an ALLC–DISC mixture under Simple Standing. The ALLD vertex remains stable in every case. Filled circles, stable equilibria; open circles, unstable equilibria; the dashed line in **a** (top) is a line of neutral equilibria. Abrupt changes in direction near the DISC vertex in the middle row of **b** and **c** arise where the high-reputation state of discriminators ends at a fold (Extended Data Fig. 4). Payoffs at each composition use private reputations relaxed from initially good records; $b/c = 5$, $e_1 = e_2 = 0.02$.

Figure asset `figures/Fig4.pdf` is provided in the companion source archive.

Figure 4. Cooperation depends on how inertia is allocated, not simply on total depth. Mean long-run cooperation $\bar{\rho}$ for every allocation of a total depth $K = k_1 + k_2 \leq 8$ between warning (k_1) and recovery (k_2) states under Scoring (**a**), Shunning (**b**), Stern Judging (**c**) and Simple Standing (**d**), with full readout ($\alpha = \beta = 1$). Points with the same total depth produce widely different outcomes. The beneficial direction is norm specific: tolerance-dominated under Scoring and Shunning, bidirectional under Stern Judging, and recovery-weighted with a non-zero warning path under Simple Standing. The black line connects the most cooperative allocation at each K, labelled (k_1, k_2). The architectures selected from reputation statistics in Fig. 2 lie on this envelope at their respective depths, although they were not chosen using cooperation. Colour shows the fraction k_1/K allocated to tolerance (blue, all tolerance; orange, all strictness; grey at $K = 0$); small horizontal offsets separate overlapping points. $\bar{\rho}$ is the realized cooperation reached by the replicator dynamics, averaged over 512 initial strategy compositions sampled uniformly from the simplex; its maximum is $1 - e_1 = 0.98$. $b/c = 5$, $e_1 = e_2 = 0.02$.

Figure asset `figures/Fig5.pdf` is provided in the companion source archive.

Figure 5. Tolerance shelters partial opportunism, but a partial readout of warning states restores resistance while preserving high cooperation. A rare opportunistic discriminator (O-DISC) withholds intended help from recipients it regards as good with probability δ, independently in every interaction ($\delta = 0$, DISC; $\delta = 1$, ALLD). It invades a resident DISC population if its normalized invasion payoff $\Lambda(\delta)$ (Eq. (3)) is positive. Stern Judging; $b/c = 5$, $e_1 = e_2 = 0.02$, $\beta = 1$. **a**, $\Lambda(\delta)$ for full one-step inertia, the selected architecture $(k_1, k_2) = (4, 3)$ with full ($\alpha = 1$) and partial ($\alpha = 0.8$) warning-state readout, and two alternative mechanisms that buffer isolated negative evidence: aggregation (good image if at least one of two sampled assessments is positive) and a quantitative score (integer score bounded at ± 5, read as good if non-negative). All five exclude ALLD, but all except $(4, 3)$ with $\alpha = 0.8$ admit an interval of intermediate opportunists; open circles mark its upper end ($\delta_c = 0.008$, 0.365, 0.137 and 0.383 for one-step inertia, $(4, 3)$ with $\alpha = 1$, aggregation and quantitative scoring, respectively). The alternative mechanisms illustrate the vulnerability and are not optimized benchmarks. Curves are clipped at $\Lambda = -0.15$. **b**, Upper end δ_c of the invasion interval (top; 0 if none) and resident reputation r_{R}^*, the intended cooperation of residents (bottom), against α for $(4, 3)$. All O-DISC mutants are excluded for $\alpha_L < \alpha \leq \alpha_U$ (green), with $\alpha_L \approx 0.567$ and $\alpha_U \approx 0.804$. The lower boundary is the fold at which the high-reputation resident state disappears (open circle; Eq. (5)); the upper boundary is the onset of invasion by weak opportunism (Eq. (6)). Residents occupy the highest strictly stable reputation state.

Figure asset `figures/ED_Fig1.pdf` is provided in the companion source archive.

Extended Data Figure 1. Replicated simulations confirm the selected architectures; their exact depths depend on the recognition criteria, but their directional pattern does not. a–d, Separation Δg against mean disagreement $\bar{d}$ (Fig. 2) for all 49 depth pairs $0 \le k_1, k_2 \le 6$ under Scoring (**a**), Shunning (**b**), Stern Judging (**c**) and Simple Standing (**d**); colour gives the total depth $K = k_1 + k_2$. Each point is the mean of 30 independent simulations ($N = 150$, equal strategy numbers, $\alpha = \beta = 1$, records initially good, $e_1 = e_2 = 0.02$). Shading marks $\Delta g \ge 0.8$ and $\bar{d} \le 0.1$. Stars mark the qualifying allocation with minimal K (ties resolved by larger Δg, then smaller $\bar{d}$): $(2,0)$, $\Delta g = 0.919$, $\bar{d} = 0.064$; $(3,0)$, 0.906, 0.069; $(4,3)$, 0.881, 0.081; $(1,2)$, 0.917, 0.056. Bootstrap 95% intervals are narrower than ± 0.001. Insets show all pairs; crosses mark no inertia. **e–h**, Allocation selected under alternative criteria for the maximal disagreement (horizontal) and minimal separation (vertical); cell colour gives its total depth. Gold outlines mark the default criteria; dark outlines enclose criteria selecting the same allocation as the default.

Figure asset `figures/ED_Fig2.pdf` is provided in the companion source archive.

Extended Data Figure 2. One-step inertia makes reputations and cooperation depend on reputational history, whereas the selected architectures are robust across contrasting initial reputations. a–d, Reputation dynamics of a homogeneous DISC population under Scoring (**a**), Shunning (**b**), Stern Judging (**c**) and Simple Standing (**d**): drift $\dot{r} = \mathscr{R}[u_{\mathrm{R}}(r)] - r$ of the probability r that a discriminator is regarded as good, without inertia (grey), with full one-step inertia (blue) and with the selected multi-step architecture (orange; depths indicated, $\alpha = \beta = 1$). Phase lines above each panel show the direction of the drift; filled and open circles mark stable and unstable reputation states. One-step inertia makes Scoring, Shunning and Stern Judging bistable; each selected architecture has a single high-reputation state. **e**, Mean long-run cooperation $\bar{\rho}$ when all private records start good (solid bars) or bad (hatched bars) and reputations then follow the strategy dynamics; 512 initial strategy compositions as in Fig. 4. One-step cooperation under Shunning and Stern Judging disappears from bad initial records; the selected architectures give identical outcomes under Scoring, Shunning and Simple Standing and nearly identical outcomes under Stern Judging. $b/c = 5$ (**e**), $e_1 = e_2 = 0.02$.

Figure asset `figures/ED_Fig9.pdf` is provided in the companion source archive.

Extended Data Figure 3. Both the readout and the depths of inertia reorganize the reputation levels and multistability of a discriminator population. Colour shows the highest stable reputation r^* of a homogeneous DISC population; hatching marks parameters with two stable reputation states, for which the colour shown is not reached from every initial reputation. **a**–**d**, One-step depths ($k_1 = k_2 = 1$) across tolerance strength α and strictness strength β under Scoring (**a**), Shunning (**b**), Stern Judging (**c**) and Simple Standing (**d**). **e**–**h**, Full readout ($\alpha = \beta = 1$) across depths $0 \leq k_1, k_2 \leq 8$. Squares mark no inertia (black) and one-step inertia (blue); stars mark the selected architectures, which lie in regions with a single high-reputation state. Long recovery paths relative to the warning path leave only a low-reputation state. $e_1 = e_2 = 0.02$; reputations do not depend on b/c.

Figure asset `figures/ED_Fig8.pdf` is provided in the companion source archive.

Extended Data Figure 4. Directional inertia changes how the reputation of discriminators erodes as defectors spread, moving the composition at which discriminators outperform defectors. All self-consistent reputation states on the ALLD–DISC edge of the simplex ($f_{\mathrm{ALLC}} = 0$) without inertia (grey), with full one-step inertia (blue) and with the selected multi-step architectures (orange; $\alpha = \beta = 1$). **a**–**d**, Reputation r^*_{DISC} of discriminators against their frequency under Scoring (**a**), Shunning (**b**), Stern Judging (**c**) and Simple Standing (**d**); solid, stable; dashed, unstable; diamonds, folds. **e**–**h**, Normalized payoff advantage of DISC over ALLD, $(\Pi_{\mathrm{DISC}} - \Pi_{\mathrm{ALLD}})/[c(1-e_1)] = (b/c) f_{\mathrm{DISC}}(r^*_{\mathrm{DISC}} - r^*_{\mathrm{ALLD}}) - \bar{r}^*$, on the state reached from initially good records. Circles mark edge equilibria, where the advantage changes sign continuously; triangles mark sign changes caused by a fold. Under Shunning and Stern Judging, one-step inertia sustains a high-reputation state only when discriminators are common ($f_{\mathrm{DISC}} \geq 0.97$ and ≥ 0.77); the selected architectures displace this fold to $f_{\mathrm{DISC}} \approx 0.51$ (Shunning, with a hysteresis loop) or remove it, leaving a smooth threshold at 0.28 (Stern Judging). Under Scoring and Simple Standing, the threshold shifts smoothly, from 0.21 to 0.08 and from 0.36 to 0.21, respectively. $b/c = 5$, $e_1 = e_2 = 0.02$.

Figure asset `figures/ED_Fig4.pdf` is provided in the companion source archive.

Extended Data Figure 5. The cooperative advantage of the selected architectures persists across benefit-to-cost ratios. Mean long-run cooperation $\bar{\rho}$ against b/c under Scoring (**a**), Shunning (**b**), Stern Judging (**c**) and Simple Standing (**d**) without inertia (grey), with full one-step inertia (blue) and with the selected multi-step architectures (orange; $\alpha = \beta = 1$). Lines interpolate means over eight independent samples of 512 initial compositions at 45 ratios between 1.001 and 10; shading, ± 1 s.e.m. across samples. Solid segments indicate that a discriminator population resists both rare ALLD and rare ALLC ($R_L < b/c < R_U$, Eq. (2)); dashed segments indicate that it does not. Labels give thresholds within the plotted range; arrows indicate $R_U > 10$. Without inertia, no ratio protects a discriminator population. Under Simple Standing, the selected architecture loses protection against ALLC above $R_U \approx 4.0$, yet mean cooperation continues to increase because the cooperative attractor is an ALLC–DISC mixture. $e_1 = e_2 = 0.02$.

Figure asset `figures/ED_Fig5.pdf` is provided in the companion source archive.

Extended Data Figure 6. Under Scoring, Shunning and Stern Judging, the selected architectures protect a discriminator population against unconditional mutants over a broad range of implementation and assessment errors. Local invasion of a resident DISC population by rare ALLD and ALLC across implementation error e_1 and assessment error e_2 (241×241 grid) under Scoring, Shunning, Stern Judging and Simple Standing (columns), without inertia (**a**–**d**), with full one-step inertia (**e**–**h**) and with the selected multi-step architectures (**i**–**l**; $\alpha = \beta = 1$). Pink, ALLD is not excluded (including neutral cases); purple, ALLD is excluded but ALLC invades; green, the resident resists both. Dashed and solid white curves mark zero invasion payoff of ALLD and ALLC, respectively. The protected region covers 60%, 67% and 31% of the plane under Scoring, Shunning and Stern Judging with multi-step inertia, at most 1.6% with one-step inertia, and only the error-free corner without inertia; Simple Standing is an exception. Residents are reached from initially good records; along $e_2 = 0$ under Scoring and Shunning this is the all-bad state, against which ALLD is neutral. This is a local invasion analysis; mean cooperation was not computed across errors. $b/c = 5$.

Figure asset `figures/ED_Fig6.pdf` is provided in the companion source archive.

Extended Data Figure 7. The full depth plane confirms that the direction of inertia organizes cooperation. Mean long-run cooperation $\bar{\rho}$ (colour; computed as in Fig. 4) for all depth pairs $0 \leq k_1, k_2 \leq 7$ under Scoring (**a**), Shunning (**b**), Stern Judging (**c**) and Simple Standing (**d**), with full readout ($\alpha = \beta = 1$). The norm-specific allocation patterns of Fig. 4 extend across the plane; the most cooperative pairs are $(5,0)$, $(7,0)$, $(7,6)$ and $(2,7)$, respectively. Dots mark pairs at which a discriminator population resists both rare ALLD and rare ALLC; under every norm these are the pairs with $k_1 \geq 2$, together with $(1,0)$ and $(1,1)$. Local protection and mean cooperation need not coincide: under Simple Standing, pairs with $k_1 = 1$ and $k_2 \geq 2$ are highly cooperative although ALLC can invade a discriminator population. $b/c = 5$, $e_1 = e_2 = 0.02$.

Figure asset `figures/ED_Fig3.pdf` is provided in the companion source archive.

Extended Data Figure 8. Directionality also operates through readout: tolerance-only readout supports discriminators and cooperation where strictness-only readout does not, with a norm-dependent response. All panels use one-step depths ($k_1 = k_2 = 1$) and vary the readout parameters α (tolerance strength) and β (strictness strength). **a–d**, Representative replicator dynamics, as in Fig. 3, under Scoring with tolerance-only readout, $(\alpha, \beta) = (1, 0)$ (**a**), and strictness-only readout, $(0, 1)$ (**b**), and under Stern Judging with tolerance-only (**c**) and strictness-only (**d**) readout. With tolerance-only readout, a discriminator population is a strict local attractor; with strictness-only readout it is not, under any of the four norms (Supplementary Information). Filled circles, equilibria stable along the boundary on which they lie; open circles, unstable equilibria. **e–h**, Mean long-run cooperation $\bar{\rho}$ across the readout plane under Scoring (**e**), Shunning (**f**), Stern Judging (**g**) and Simple Standing (**h**) (41×41 grid; 512 initial compositions per point; non-linear colour scale). Cooperation is highest at tolerance-only readout and vanishes at strictness-only readout under Scoring, Shunning and Stern Judging, but remains high across the plane under Simple Standing. **i–l**, Local invasion of a resident DISC population by rare ALLD and ALLC under the same norms (801×801 grid), with colours and curves as in Extended Data Fig. 6. $b/c = 5$, $e_1 = e_2 = 0.02$.

Figure asset `figures/ED_Fig7.pdf` is provided in the companion source archive.

Extended Data Figure 9. Protective readout intervals exist across norms and benefit-to-cost ratios, but their location depends on the norm and the inertia architecture. Security of a resident DISC population against the complete IID O-DISC family ($0 < \delta \leq 1$; Fig. 5) for full one-step inertia and the selected multi-step architectures ($\beta = 1$). **a**, Tolerance strength α at $b/c = 5$. Colour shows the resident reputation r_R^*, the intended cooperation of residents; black outlines enclose secure intervals. Open circles mark interval ends at folds of the high-reputation resident state, which are excluded. Selected multi-step intervals are $[0, 0.807]$ (Scoring), $(0.591, 0.806]$ (Shunning), $(0.567, 0.804]$ (Stern Judging) and $[0, 1]$ (Simple Standing); one-step intervals are $(0.432, 0.967]$, $(0.865, 0.990]$, $(0.852, 0.989]$ and $[0, 0.965]$. **b**–**e**, Secure regions (black outlines) across α and $b/c \in [2, 8]$ under Scoring (**b**), Shunning (**c**), Stern Judging (**d**) and Simple Standing (**e**) for one-step (top) and selected multi-step (bottom) inertia; colour, r_R^*, which does not depend on b/c. Lower boundaries set by folds (Eq. (5)) do not depend on b/c, whereas upper boundaries (Eq. (6)) rise with b/c. Circles mark $(\alpha, b/c) = (0.8, 5)$, which is secure for all four selected architectures ($r_R^* = 0.978$–0.990) but not for one-step Shunning and Stern Judging. Security refers to the tested family of independent omissions only. $e_1 = e_2 = 0.02$.

Supplementary Information

To be completed following the author's outline. Detailed derivations, additional validation and supplementary analyses will be added here.